SPENGERGASSE
ausbildung mit zukunft

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Klasse: 6ACIF		**Raum Nr.:** **Platz Nr.:**
Lehrer: Prof. Spanring, Prof. Horny	**Abgabedatum: 17.03.2025**	**Bewertung:**

LAB2 - DHCP4+Relay

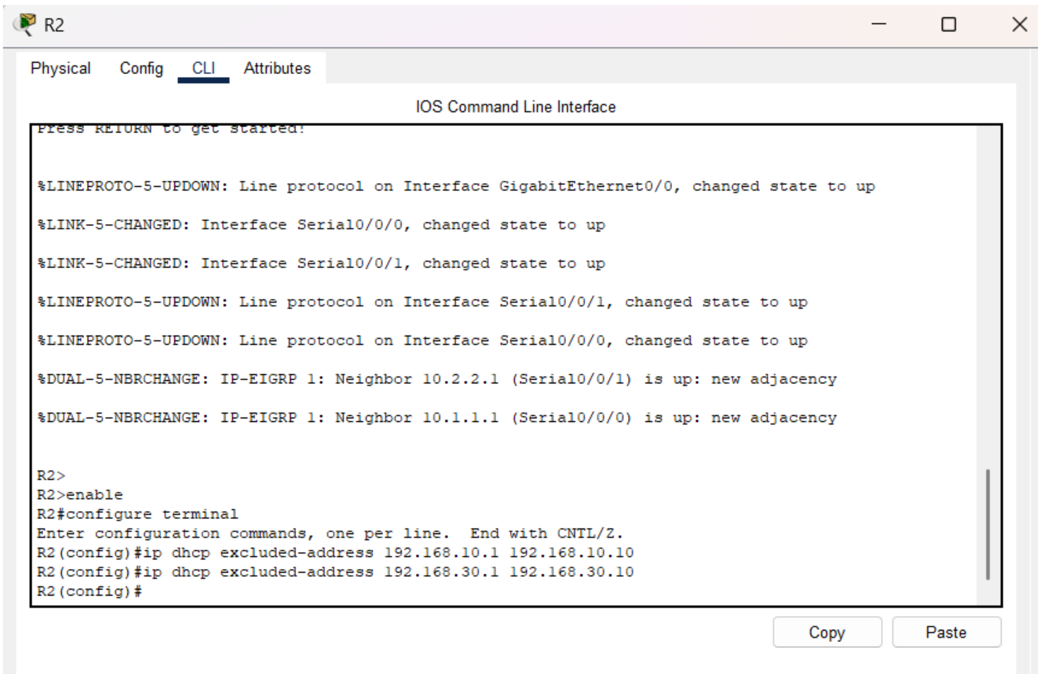
- **Aufgabenstellung**

  Packet Tracer - Configuring DHCP Using Cisco IOS
  Ziel dieser Aufgabe ist es, einen Cisco-Router als DHCP-Server zu konfigurieren, DHCP-Relay einzurichten und die DHCP-Funktionalität durch verschiedene Tests zu überprüfen.

- **Durchführungsbeschreibung (Arbeitsschritte)**

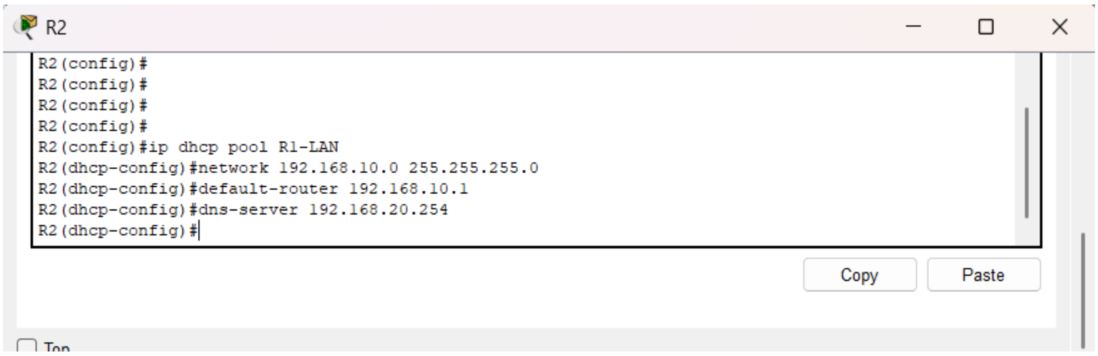
Teil 1: Konfigurieren eines Routers als DHCP-Server

Schritt 1: Konfigurieren der ausgeschlossenen IPv4-Adressen
1. Den CLI-Modus des Routers R2 aufrufen.
2. Die folgenden Befehle eingeben, um die ersten 10 Adressen aus den R1- und R3-LANs auszuschließen:



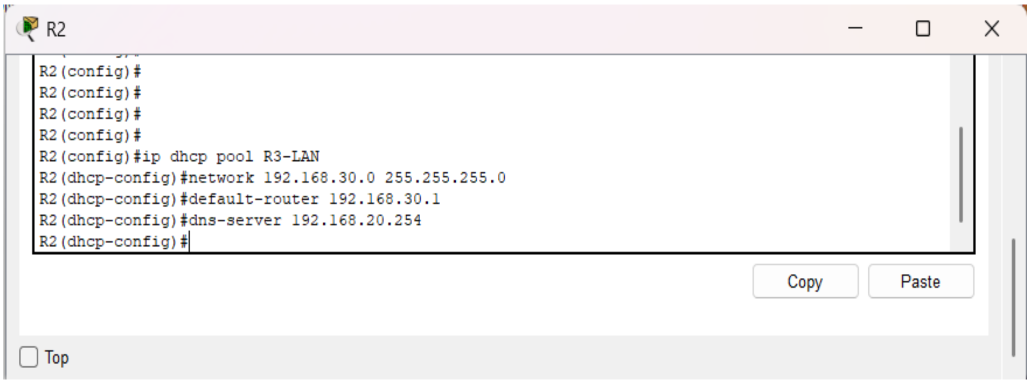
Schritt 2: Erstellen eines DHCP-Pools für das R1-LAN

1. Einen DHCP-Pool mit dem Namen R1-LAN erstellen:



Schritt 3: Erstellen eines DHCP-Pools für das R3-LAN

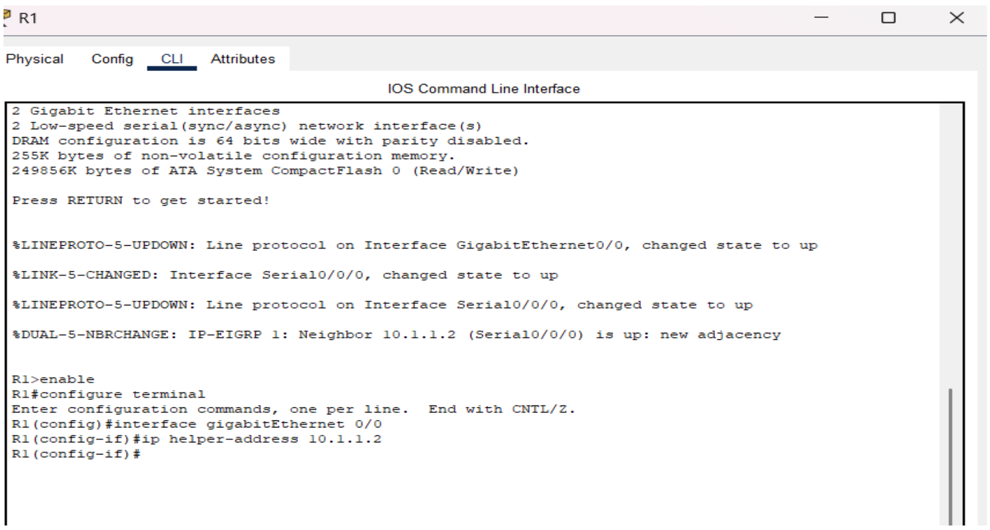
1. Einen weiteren DHCP-Pool mit dem Namen R3-LAN erstellen:

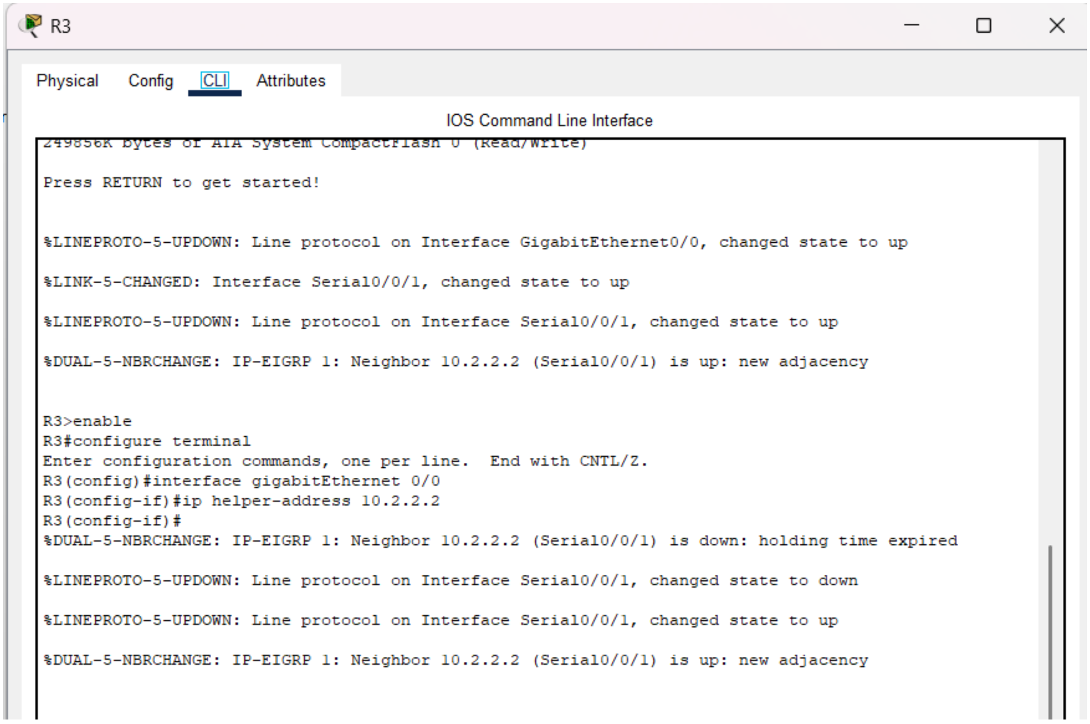


Teil 2: Konfigurieren des DHCP-Relay

Schritt 1: Konfigurieren von R1 und R3 als DHCP-Relay-Agenten

1. Auf R1 und R3 in die Konfiguration der Schnittstelle GigabitEthernet0/0 wechseln:

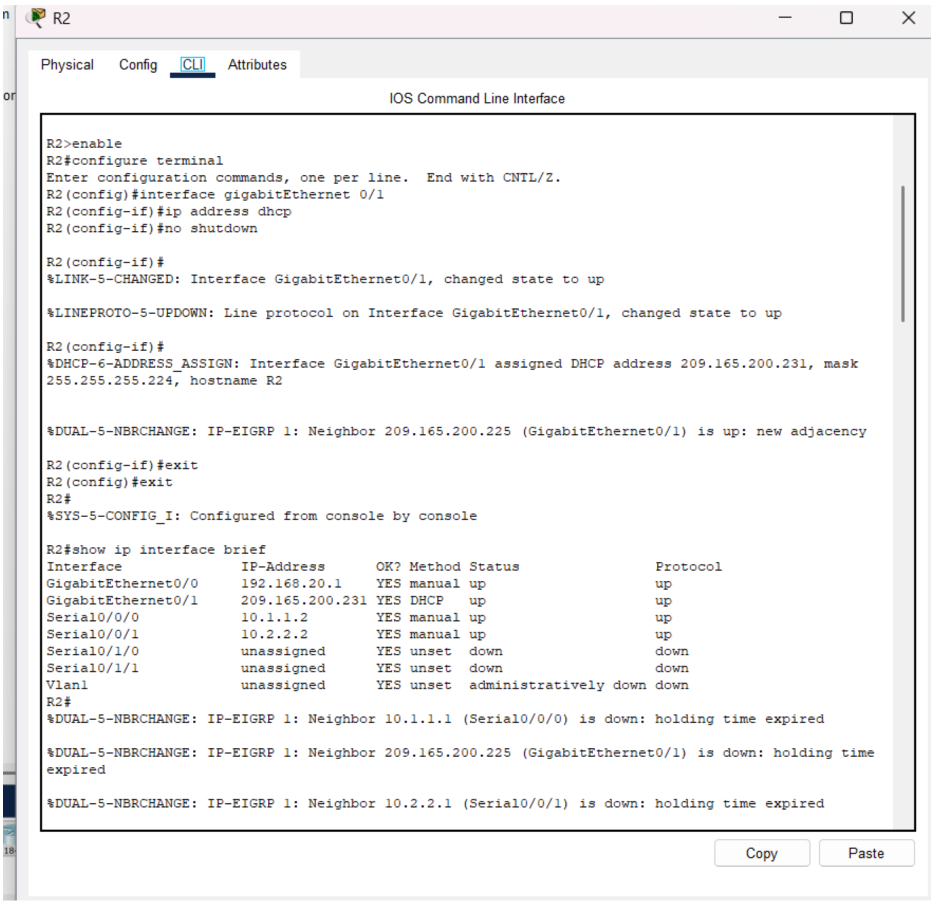





```
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!


%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up

%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 10.2.2.2 (Serial0/0/1) is up: new adjacency


R3>enable
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#interface gigabitEthernet 0/0
R3(config-if)#ip helper-address 10.2.2.2
R3(config-if)#
%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 10.2.2.2 (Serial0/0/1) is down: holding time expired

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up

%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 10.2.2.2 (Serial0/0/1) is up: new adjacency
```

Teil 3: Konfigurieren von R2 als DHCP-Client

1. Schnittstelle GigabitEthernet0/1 von R2 auf DHCP setzen:
2. Überprüfung, ob R2 eine IP-Adresse erhalten hat:

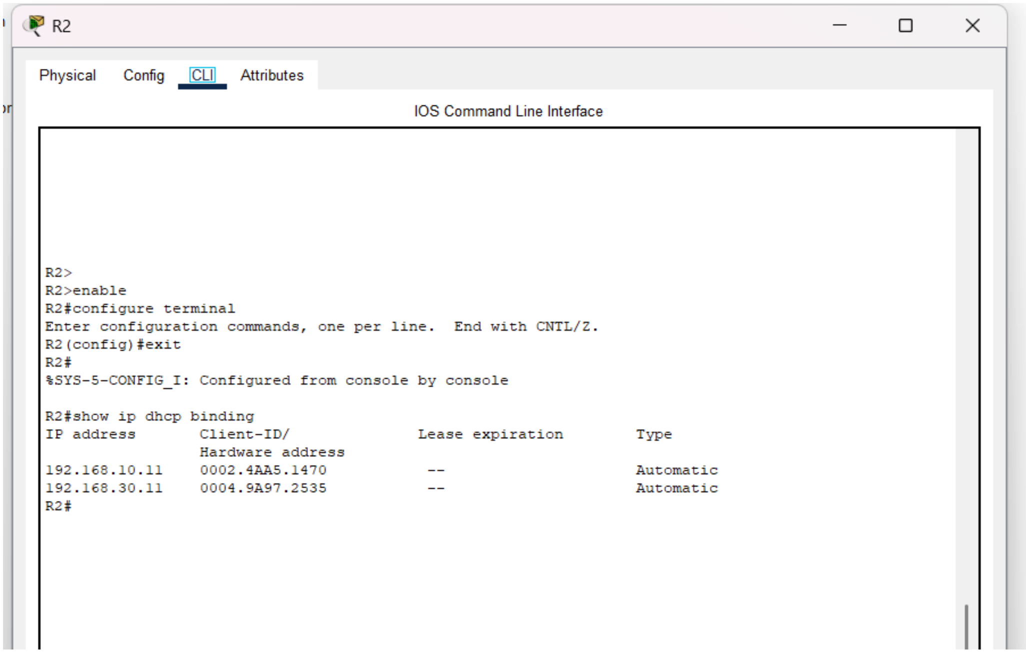



```
R2>enable
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#interface gigabitEthernet 0/1
R2(config-if)#ip address dhcp
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R2(config-if)#
%DHCP-6-ADDRESS_ASSIGN: Interface GigabitEthernet0/1 assigned DHCP address 209.165.200.231, mask
255.255.255.224, hostname R2


%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 209.165.200.225 (GigabitEthernet0/1) is up: new adjacency

R2(config-if)#exit
R2(config)#exit
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.20.1    YES manual up                    up
GigabitEthernet0/1     209.165.200.231 YES DHCP   up                    up
Serial0/0/0            10.1.1.2        YES manual up                    up
Serial0/0/1            10.2.2.2        YES manual up                    up
Serial0/1/0            unassigned      YES unset  down                  down
Serial0/1/1            unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
R2#
%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 10.1.1.1 (Serial0/0/0) is down: holding time expired

%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 209.165.200.225 (GigabitEthernet0/1) is down: holding time
expired

%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 10.2.2.1 (Serial0/0/1) is down: holding time expired
```

Teil 4: Überprüfung von DHCP und Konnektivität

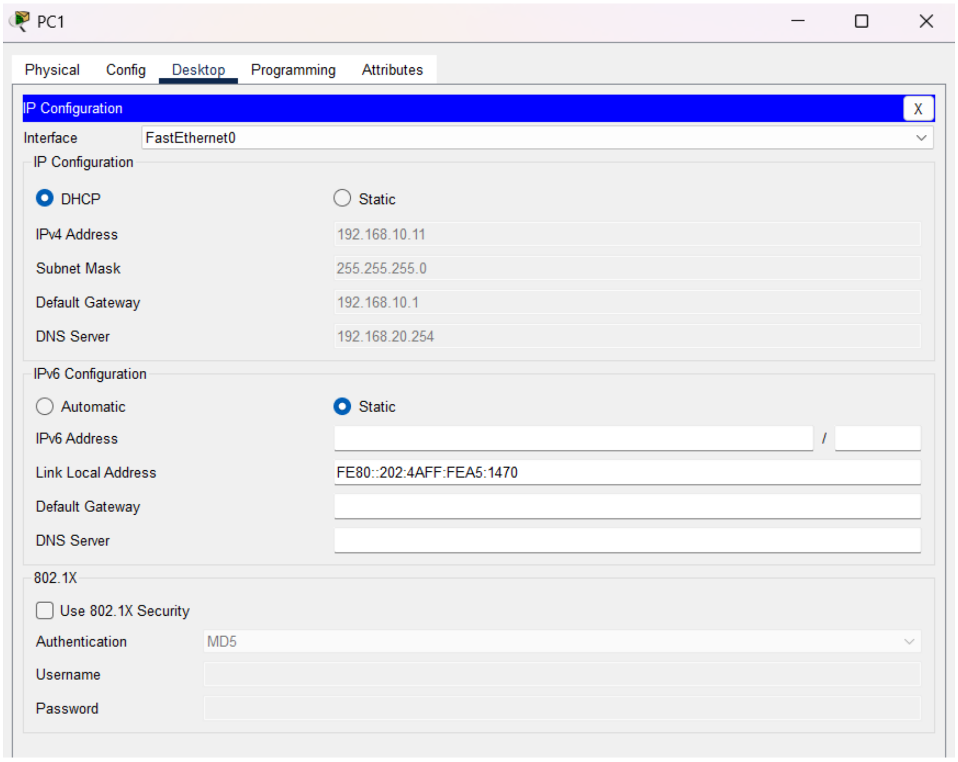
Schritt 1: Überprüfung der DHCP-Bindungen

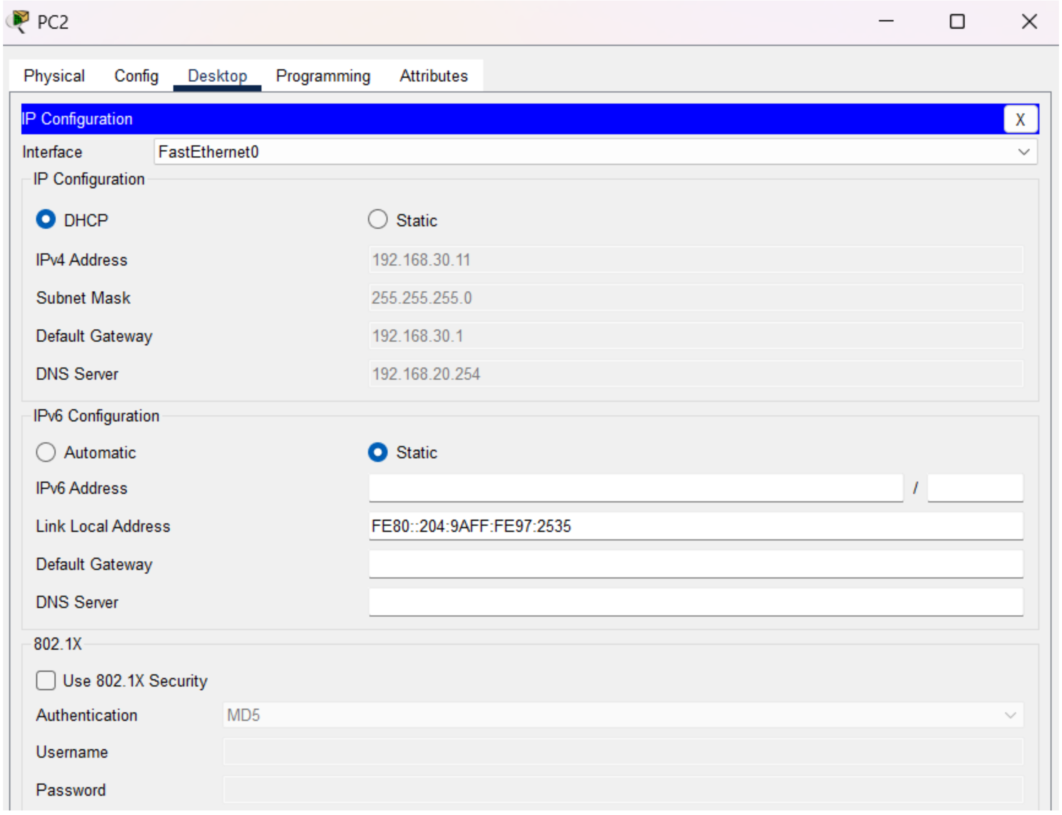
1. Auf R2 überprüfen, welche IP-Adressen über DHCP vergeben wurden:

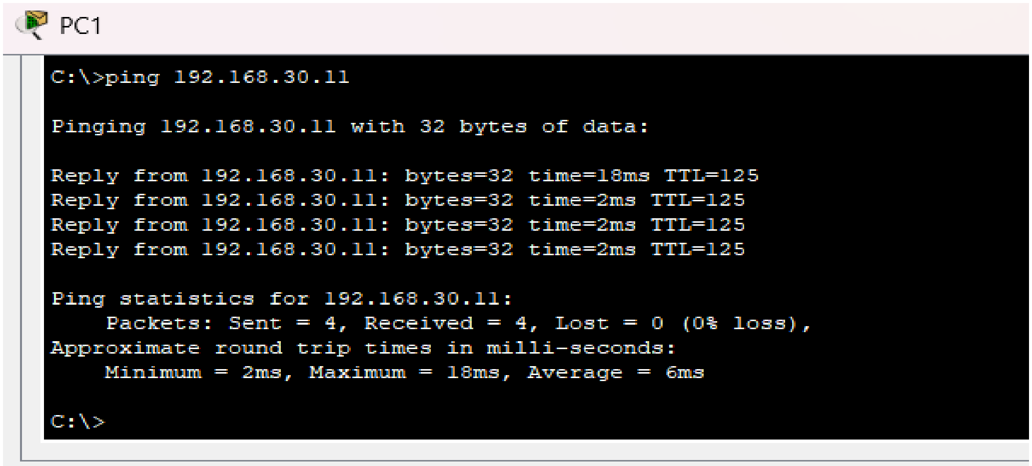


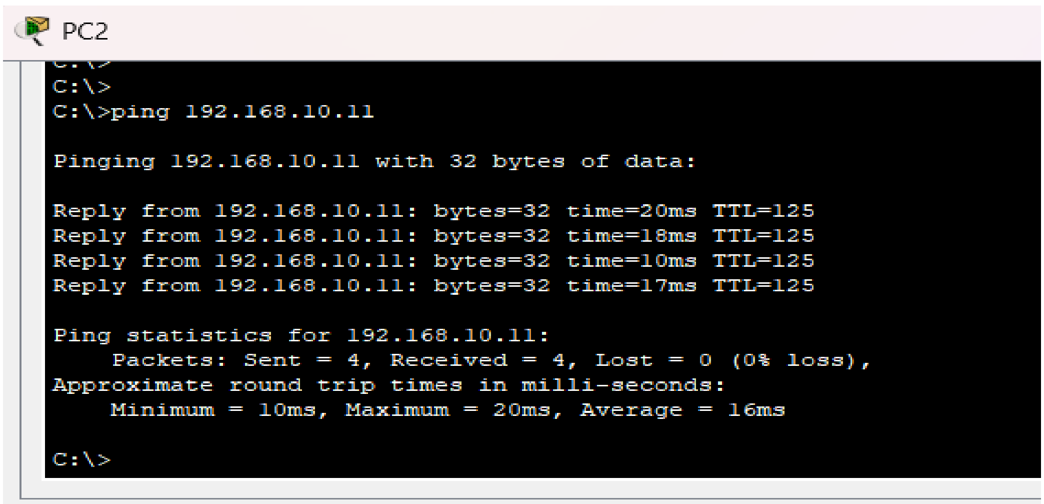
Schritt 2: Überprüfung der Konnektivität

1. Auf PC1 und PC2 sicherstellen, dass sie auf DHCP eingestellt sind.
2. Falls nötig, erneuern der DHCP-Lease auf PC1 und PC2.
3. Überprüfen, ob PC1 und PC2 sich gegenseitig anpingen können:




PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.30.11
Subnet Mask
255.255.255.0
Default Gateway
192.168.30.1
DNS Server
192.168.20.254
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::204:9AFF:FE97:2535
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password


PC1
C:\>ping 192.168.30.11
Pinging 192.168.30.11 with 32 bytes of data:
Reply from 192.168.30.11: bytes=32 time=18ms TTL=125
Reply from 192.168.30.11: bytes=32 time=2ms TTL=125
Reply from 192.168.30.11: bytes=32 time=2ms TTL=125
Reply from 192.168.30.11: bytes=32 time=2ms TTL=125
Ping statistics for 192.168.30.11:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 18ms, Average = 6ms
C:\>


PC2
C:\>
C:\>ping 192.168.10.11
Pinging 192.168.10.11 with 32 bytes of data:
Reply from 192.168.10.11: bytes=32 time=20ms TTL=125
Reply from 192.168.10.11: bytes=32 time=18ms TTL=125
Reply from 192.168.10.11: bytes=32 time=10ms TTL=125
Reply from 192.168.10.11: bytes=32 time=17ms TTL=125
Ping statistics for 192.168.10.11:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 10ms, Maximum = 20ms, Average = 16ms
C:\>

Ping von PC2 zum DNS-Server (192.168.20.254):

PC2

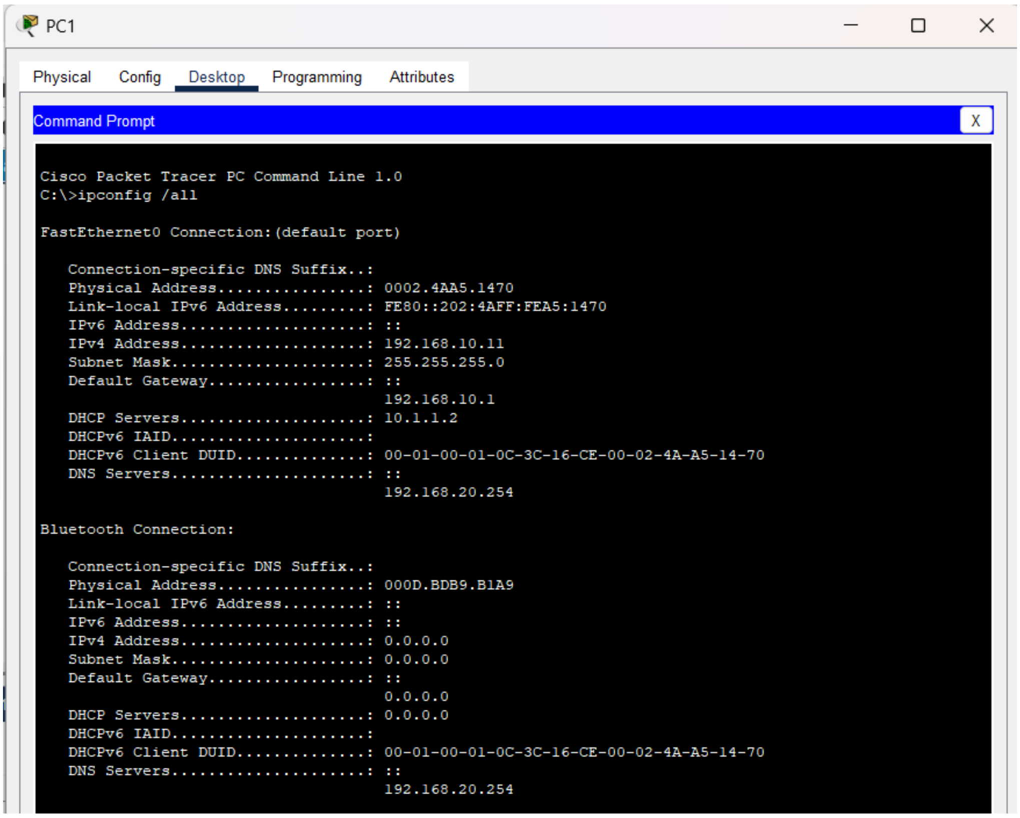
```
C:\>ping 192.168.20.254

Pinging 192.168.20.254 with 32 bytes of data:

Reply from 192.168.20.254: bytes=32 time=10ms TTL=126
Reply from 192.168.20.254: bytes=32 time=1ms TTL=126
Reply from 192.168.20.254: bytes=32 time=1ms TTL=126
Reply from 192.168.20.254: bytes=32 time=14ms TTL=126

Ping statistics for 192.168.20.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 14ms, Average = 6ms

C:\>
```

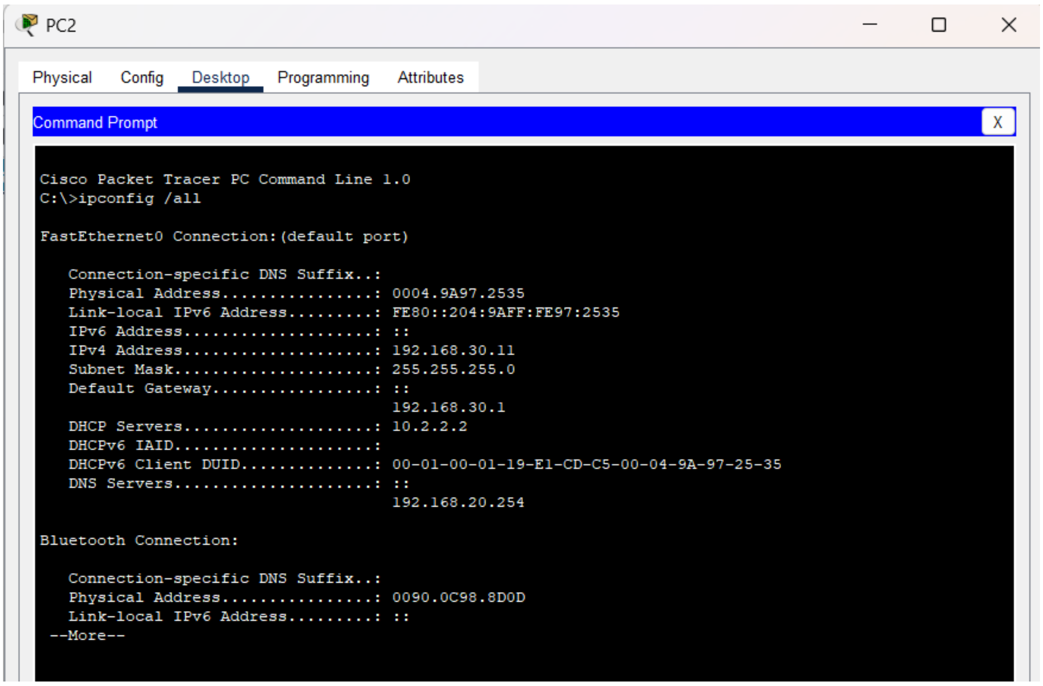
Ping von PC1 zu einer externen IP-Adresse (209.165.202.158):

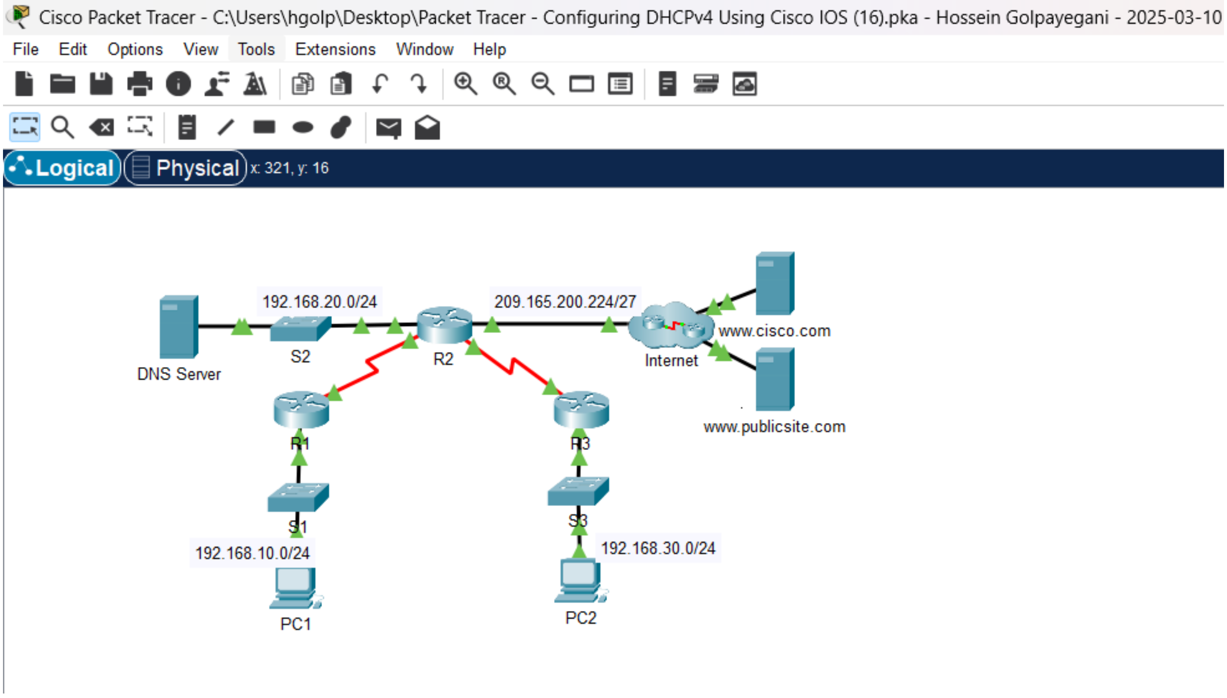
PC1

```
C:\>
C:\>
C:\>ping 209.165.202.158

Pinging 209.165.202.158 with 32 bytes of data:

Reply from 209.165.202.158: bytes=32 time=12ms TTL=125
Reply from 209.165.202.158: bytes=32 time=1ms TTL=125
Reply from 209.165.202.158: bytes=32 time=1ms TTL=125
Reply from 209.165.202.158: bytes=32 time=2ms TTL=125

Ping statistics for 209.165.202.158:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 12ms, Average = 4ms

C:\>
```

Überprüfung der IP-Konfiguration auf den Clients (PC1 und PC2)







Packet Tracer - Configuring DHCP Using Cisco IOS

Addressing Table

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
	G0/0	192.168.10.1	255.255.255.0	N/A
R1	S0/0/0	10.1.1.1	255.255.255.252	N/A
	G0/0	192.168.20.1	255.255.255.0	N/A
	G0/1	DHCP Assigned	DHCP Assigned	N/A
	S0/0/0	10.1.1.2	255.255.255.252	N/A
R2	S0/0/1	10.2.2.2	255.255.255.252	N/A
	G0/0	192.168.30.1	255.255.255.0	N/A
R3	S0/0/1	10.2.2.1	255.255.255.0	N/A
PC1	NIC	DHCP Assigned	DHCP Assigned	DHCP Assigned
PC2	NIC	DHCP Assigned	DHCP Assigned	DHCP Assigned
DNS Server	NIC	192.168.20.254	255.255.255.0	192.168.20.1

Objectives

Part 1: Configure a Router as a DHCP Server

Part 2: Configure DHCP Relay

Part 3: Configure a Router as a DHCP Client

Part 4: Verify DHCP and Connectivity

Scenario

Time Elapsed: 05:35:11 Completion: 100/100

Top Dock Check Results Back 1/1 Next

Cisco Packet Tracer - C:\Users\hgolp\Desktop\Packet Tracer - Configuring DHCPv4 Using Cisco IOS (16).pka -...

File Edit Options View Tools Extensions Window Help

Activity Results Time Elapsed: 02:40:44

Congratulations Hossein Golpayegani! You completed the activity.

Overall Feedback | Assessment Items | Connectivity Tests

Congratulations! You successfully completed the **Packet Tracer - Configuring DHCPv4 Using Cisco IOS** activity.